Wiley Analytical Science

# Automating the Laboratory: Maximizing Efficiency Through Smart Solutions

## April 27, 9:00am - 10:00am EDT

Automation has reached ever larger areas recently. Due to the development of numerous new, cost-effective automation devices and modules, the automation of laboratory processes is also of increasing interest to small and medium-sized companies and research laboratories. Different degrees of automation from the automation of individual sub-steps to partial automation and finally full automation without manual intervention by an operator are possible.

**Watch this session during the WAS Virtual Conference:**

**Prof. Kerstin Thurow**

**Prof. Thole Züchner**

**Register Now**

This talk is sponsored by SARTORIUS

Check for updates

Received: 9 February 2023 | Revised: 15 March 2023 | Accepted: 16 March 2023

DOI: 10.1002/dta.3473

RESEARCH ARTICLE

WILEY

# Myostatin inhibitory peptides in sports drug testing

Katja Walpurgis[1] | Johannes Agricola[2] | Andreas Thomas[1] | Mario Thevis[1,3]

[1]Institute of Biochemistry/Center for Preventive Doping Research, German Sport University Cologne, Cologne, Germany

[2]Division of BioAnalytical Chemistry, Faculty of Science, Vrije Universiteit Amsterdam, Amsterdam, Netherlands

[3]European Monitoring Center for Emerging Doping Agents (EuMoCEDA), Cologne/Bonn, Germany

**Correspondence**
Katja Walpurgis, Institute of Biochemistry/Center for Preventive Doping Research, German Sport University Cologne, Cologne, Germany.
Email: k.walpurgis@biochem.dshs-koeln.de

**Funding information**
National Anti-Doping Agency (NADA) Germany; Federal Ministry of the Interior, Building, and Community of the Federal Republic of Germany; ERASMUS+ Programme of the European Commission; Manfred-Donike Institute for Doping Analysis

## Abstract

Across species, skeletal muscle mass is negatively regulated by the TGF-β cytokine myostatin (MSTN). Inhibitors of this growth factor and its signaling pathways are therefore not only promising therapeutics for muscular diseases but also potential performance-enhancing agents in sports. Within this study, protein precipitation and liquid chromatography high-resolution tandem mass spectrometry (LC-HRMS/MS) were employed to develop a detection method for six novel MSTN inhibitory peptides derived from the regulatory MSTN propeptide and the natural MSTN inhibitor follistatin (FST) from doping control serum samples. The approach was comprehensively characterized and found to allow for a specific detection down to concentrations of 3–9 ng/mL. Moreover, several potential metabolites of the drug candidates referred to as DF-3, DF-25, and Peptide 7 were identified as valuable complementary analytical targets for doping control analytical assays. Overall, the acquired data pave the way for an implementation of MSTN inhibitory peptides into routine sports drug testing. Even though no drug candidate has obtained clinical approval yet, a proactive development of detection assays is of utmost importance to deter athletes from misusing such compounds, which are readily available for research purposes and on the black market.

**KEYWORDS**
doping, LC-HRMS/MS, myostatin inhibitor, sports

## 1 | INTRODUCTION

Skeletal muscle mass is negatively regulated by cytokines of the transforming growth factor β (TGF-β) superfamily, and especially, the role of myostatin (MSTN) has been extensively studied in the past. The disulfide-linked dimer has a molecular mass of 25 kDa, is predominantly expressed in developing and adult skeletal muscle, and is highly conserved across species.[1] Similar to other TGF-β cytokines, MSTN is synthesized as precursor polypeptide comprising an N-terminal signal sequence for secretion, a propeptide with regulatory functions, and a C-terminal bioactive fragment.[1] Following cellular secretion, proteolytic cleavage, and dimerization, MSTN is kept in an inactive latent complex by two non-covalently bound propeptide molecules and several other proteins such as follistatin (FST).[2] Proteolysis results in the release of the C-terminal MSTN dimer, which exerts its biological role through binding to activin type II serine/threonine kinase transmembrane receptors (ActRIIA & -B), which is followed by the activation of Smad proteins functioning as intracellular signal transducer to finally regulate gene expression.[2,3]

As MSTN knockout mice were found to be characterized by a significantly elevated muscle mass due to an increased number and size of muscle fibers,[1] the MSTN signaling pathway represents a promising therapeutic target for the treatment of muscular diseases.[2,3] Different strategies for its inhibition have been developed, comprising targeted

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.
© 2023 The Authors. *Drug Testing and Analysis* published by John Wiley & Sons Ltd.

(e.g., anti-MSTN antibodies) as well as multi-targeting approaches (e.g., receptor competitors and anti-ActRII antibodies).[4] So far, no drug candidate has obtained clinical approval, which can—to some extent—be explained by the fact that especially, agents directly targeting MSTN were found to be less effective in humans than in animal models.[5] A possible explanation for this phenomenon could be differing serum levels of MSTN and other negative regulators of muscle mass between species.[5] Moreover, unexpected side effects such as epistaxis and telangiectasias were observed for agents that also bind to other TGF-β cytokines such as the receptor competitor ACE-031 (ActRIIB-Fc).[4,6]

Besides the abovementioned protein-based MSTN inhibitors, several peptidic drug candidates have been designed. Takayama et al.[7] identified the minimum amino acid sequence of the murine MSTN propeptide, which can effectively inhibit the cytokine both in vitro and in vivo. Subsequently, the key peptide sequence was modified by N-terminal acetylation,[8] amino acid exchanges,[9] shortening,[10] and the introduction of non-natural amino acids[10] to further increase the inhibitory activity of the peptide (Table 1). Furthermore, Saitoh et al.[11,12] described MSTN-specific peptides derived from the N-terminal MSTN-binding domain of FST, whose effectiveness was optimized by different amino acid substitutions.

Protein- and peptide-based MSTN inhibitors are not only potential therapeutics but can also be considered as performance-enhancing agents in sports. Their misuse is prohibited both in- and out-of-competition, and they are listed in section S4 (Hormone and Metabolic Modulators) of the World Anti-Doping Agency's (WADA's) Prohibited List.[13] For several protein-based drug candidates such as the MSTN-neutralizing antibodies MYO-029 and Domagrozumab,[14,15] the anti-ActRII antibody Bimagrumab,[16] and inhibitors derived from FST,[17,18] doping control detection methods have been proactively developed. This is of utmost importance, as, even without clinical approval, reference material for such emerging drugs is often available for research purposes and/or distributed on the black market. Therefore, the aim of this research project was to develop the first doping control detection assay for exemplary MSTN inhibitory peptides from serum by using LC-HRMS/MS.

## 2 | MATERIAL AND METHODS

### 2.1 | Reference material

Based on the amino acid sequences described in the literature,[7,9–12] six different MSTN inhibitory peptides derived from the MSTN propeptide and FST (Table 1) were custom-synthesized by GenScript (Leiden, Netherlands). Additionally, two internal standard (ISTD) peptides with amino acid exchanges (Ala → Gly and Gly → Ala; Table 1) were designed to monitor the performances of sample preparation and analysis.

### 2.2 | Samples

For method development and characterization, human serum obtained from male AB plasma was purchased from Sigma-Aldrich (Munich, Germany). Additionally, serum and plasma samples were collected from 10 healthy volunteers (5 male and 5 female) to determine certain validation parameters such as specificity and robustness. All participants provided written informed consent, and the study was also approved by the local ethical committee (DSHS No. 139/2021).

### 2.3 | Sample preparation

The protocol used for sample extraction was adapted from earlier studies[19] as follows: 100 µL of serum/plasma were mixed with the ISTD at a concentration of 25 ng/mL as well as 100 µL of 1% acetic acid (HAc), and 400 µL of ice-cold acetonitrile (ACN) were added to achieve protein precipitation. Following centrifugation (7 min, 13,000×*g*), supernatants were transferred to Protein LoBind® Tubes and evaporated at ∼50°C. Then, samples were reconstituted with 50 µL of 2% HAc, centrifuged again (10 min, 13,000×*g*), transferred to plastic vials, and subjected to LC-HRMS/MS analysis.

**TABLE 1** Details on the target analytes and ISTDs.

| Name | N-terminus | Amino acid sequence | C-terminus | Modification | Derived from … | # amino acids | Monoisotopic mass (Da) |
|---|---|---|---|---|---|---|---|
| Peptide 7 | H | WRQNTRYSRIEAIKIQILSKLRL | $NH_2$ | - | MSTN Propeptide | 23 | 2,883.70 |
| Peptide 3d | **[X]** | RQNTRYSRIE**W**IKIQI**I**SKLRL | $NH_2$ | [X] = 2-naphthyloxyacetyl | MSTN Propeptide | 23 | 2,996.72 |
| Peptide 8a | H | **WY** I**R[D-TRP]**IK**[Chg]**QI**W**SK**[Chg]**RL | $NH_2$ | [Chg] = cyclohexylglycine | MSTN Propeptide | 16 | 2,252.35 |
| DF-3 | H | VNDNTLFKWMIFNG | $NH_2$ | - | FST | 14 | 1,696.84 |
| DF-25 | H | VNDNTLFKW**[Thi]**IFNG | $NH_2$ | [Thi] = 2-thienylalanine | FST | 14 | 1,718.82 |
| DF-100 | H | **[Phg]**NDNTLFK**F[Hph]**I**[Phe(4-F)]**NG | $NH_2$ | [Phg] = α-phenylglycine<br>[Hph] = homophenylalanine<br>[Phe(4-F)] = 4-fluorophenylalanine | FST | 14 | 1,739.87 |
| ISTD1 | H | WRQNTRYSRIE**G**IKIQILSKLRL | $NH_2$ | - | MSTN Propeptide | 23 | 2,869.69 |
| ISTD2 | H | VNDNTLFKWMIFN**A** | $NH_2$ | - | FST | 14 | 1,710.86 |

19427611, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.3473 by Ohio State University Ohio Sta, Wiley Online Library on [04/04/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

## 2.4 | LC-HRMS/MS

LC-HRMS/MS analysis was conducted on an Orbitrap Exploris™ 480 mass spectrometer (Thermo Fisher Scientific, Bremen, Germany) coupled to a Vanquish™ UHPLC (Thermo Fisher Scientific, Bremen, Germany) equipped with an Accucore™ Phenyl-Hexyl trapping column (3 × 10 mm, 2.6 µm; Thermo Fisher Scientic, Bremen, Germany) and Poroshell EC-C18 column analytical column (3 × 50 mm, 2.7 µm; Agilent Technologies, Santa Clara, CA, USA). The column and autosampler temperatures were set to 30°C and 10°C. Of each sample, 10 µL were injected, and LC solvents were 0.1% formic acid with 1% dimethyl sulfoxide (DMSO) in water (A) and 0.1% formic acid with 1% DMSO in ACN (B). The LC gradient (total run time: 15 min) was run as follows with a flow rate of 400 µL/min: After trapping for 1 min with 10% B, the amount of solvent B increased to 100% within 9 min and was maintained for 1 min. Then, re-equilibration with 10% B was performed for 4 min.

The mass spectrometer was operated in positive mode with an ionization voltage of 3 kV. The temperature of the ion transfer tube was set to 320°C, and data were acquired by full MS and targeted $MS^2$ experiments. Full scan mass spectra were recorded from *m/z* 400–2,000 with a resolution of 60,000 full width at half maximum (FWHM; at *m/z* 200). Additionally, $tMS^2$ experiments were performed with an isolation window of $m/z = 2$ at a resolution of 15,000 FWHM (at *m/z* 200) and a scan range of *m/z* 200–1,500. Moreover, an inclusion list with the accurate mass-to-charge ratios for the most abundant charge states of the target peptides (Table 2) was employed. For MS/MS experiments, nitrogen obtained from a $N_2$-generator (CMC, Eschborn, Germany) was used as collision gas, and the normalized collision energy (NCE) was set to 25%. The system was calibrated according to the manufacturer's recommendations with a mixture of caffeine, the tetrapeptide MRFA (methionine-arginine- phenylalanine-alanine), and Ultramark 1,621 (Thermo Fisher Scientific).

MS data were analyzed by using Thermo Xcalibur Software (Version 4.0.27.10, 2015). For each target analyte, one precursor-product ion transition was evaluated, and a second precursor-product ion pair should be employed for confirmation purposes (Table 2).

## 2.5 | Method optimization

For peptide extraction, different protocols including both protein precipitation alone and in combination with subsequent solid-phase extraction (SPE) (HR-XA [3 mL/60 mg, Macherey-Nagel, Düren, Germany], HR-XC [3 mL/60 mg, Macherey-Nagel, Düren, Germany], and HLB [OASIS, 3ccm/60 mg, Waters, Eschborn, Germany]) were tested. Protein precipitation without subsequent SPE yielded the best results, and both the pH value of the samples and the amount of ACN used were optimized to maximize peptide recovery:

1. pH: 100 µL of serum were mixed with 100 µL of 1% HAc, deionized water, or 1% ammonia ($NH_3$) and extracted as described above. All samples were prepared in duplicate.
2. Amount of ACN: After mixing 100 µL of serum with 100 µL of 1% HAc, 200/400/600/800 µL of ACN were added for protein precipitation. Then, samples were subjected to evaporation and LC-HRMS/MS as described in Sections 2.3 and 2.4. All samples were prepared in duplicate.

## 2.6 | Method characterization

The presented approach was comprehensively characterized according to current WADA guidelines,[20] and the following parameters were determined:

- Selectivity:

  Ten different blank serum samples (five male and five female) were tested for the presence of interfering signals.

- Linearity

  Eleven serum samples were fortified with 0, 1, 2.5, 5, 7.5, 10, 20, 40, 60, 80, and 100 ng/mL of each of the six target analytes and analyzed as described above. The ISTD-normalized peak areas were used

**TABLE 2** Precursor/product ion pairs used for the mass spectrometric identification of the target peptides.

| | | | | Product ion 1 | | Product ion 2 | |
|---|---|---|---|---|---|---|---|
| Name | Monoisotopic mass (Da) | Precursor ion (*m/z*) | Charge state | ID | (*m/z*) | ID | (*m/z*) |
| Peptide 7 | 2,883.70 | 481.8 | 6 | $y_3$ | 400.30 | $y_5$ | 615.43 |
| Peptide 3d | 2,996.72 | 500.6 | 6 | $y_3$ | 400.30 | $y_5$ | 615.43 |
| Peptide 8a | 2,252.35 | 564.3 | 4 | $y_5$ | 641.45 | $y_6$ | 827.53 |
| DF-3 | 1,696.84 | 849.4 | 2 | [FN] | 262.12 | $b_{10}$ | 1249.61 |
| DF-25 | 1,718.82 | 861.0 | 2 | [FN] | 262.12 | $b_{10}$ | 1271.59 |
| DF-100 | 1,739.87 | 871.5 | 2 | [Phe(4-F)N] | 280.11 | $b_{10}$ | 1274.62 |
| ISTD1 | 2,869.69 | 479.5 | 6 | $y_3$ | 400.30 | $y_5$ | 615.43 |
| ISTD2 | 1,710.86 | 856.4 | 2 | [FN] | 262.12 | $b_{10}$ | 1249.61 |

19427611, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.3473 by Ohio State University Ohio Sta, Wiley Online Library on [04/04/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

to construct calibration curves and determine the linearity by regression analysis.

- Limit of detection (LOD)

Six serum samples were fortified with the four lowest analyte concentrations detectable during linearity measurements (2, 5, 10, and 20 ng/mL) and prepared as described in Sections 2.3 and 2.4. For LOD determination, a detection rate of >95% was applied.

- Identification capability

The identification capability of the approach was demonstrated by analyzing 10 different serum samples fortified with 20 ng/mL of the different MSTN inhibitory peptides.

- Recovery

The recovery of the method was determined at two different concentrations. For each concentration level, six blank serum samples and six specimens fortified with 20 or 40 ng/mL of the model peptides were prepared as described above. Following evaporation and reconstitution, the extracts obtained from the blank specimens were mixed with the respective amount of the different analytes. Finally, the recovery was estimated by comparing the normalized peak areas of the samples spiked before and after analyte extraction.

- Carryover

To determine the risk for analyte carryover, the extracts obtained from two blank serum samples were immediately injected after a serum specimen fortified with 80 ng/mL.

- Matrix effects

Both blank serum extracts and deionized water were fortified with the analytes at a concentration of 20 ng/mL shortly before LC-MS/MS analysis, and the measured peak areas were compared to investigate ion suppression/enhancement effects.

- Robustness

The robustness was assessed by analyzing 10 different plasma samples (instead of serum) fortified with 20 ng/mL of the six model compounds.

- Stability

Analyte stability was investigated both in unprocessed serum and sample extracts by comparing absolute peak areas detected for the different target compounds. A serum sample was fortified with 20 ng/mL of the different target analytes, and aliquots were stored for up to 14 days at RT or 4°C. At defined time points (RT: 0/1/2/4/8/24/48/72/96/168/336 h; 4°C: 0/2/4/8/24/48/96/168/336 h), the respective aliquot was extracted as described above and stored at −20°C until analysis (maximum storage time: 2 weeks). Moreover, serum sample extracts containing the target analytes at a concentration of 20 ng/mL were incubated for 140 h in the autosampler with a temperature of 10°C and subsequently analyzed.

## 2.7 | Metabolism experiments

To identify metabolites of the target analytes, an approach published by Thomas et al.[21] was adapted: A blank serum sample and six serum aliquots separately fortified with one of the MSTN inhibitory peptides at a concentration of 250 ng/mL were incubated for 2 h at RT and extracted as described in Section 2.3. Then, an untargeted LC-MS/MS analysis was conducted on the LC-HRMS system described above. A full MS experiment was performed covering the scan range of *m/z* 300–2,000 with a resolution of 60,000 FWHM (at *m/z* 200). MS/MS data were acquired by all-ion fragmentation (AIF) with a scan range from *m/z* 100–1,000, a resolution of 60,000 FWHM (at *m/z* 200), and an NCE of 25%. Subsequently, $tMS^2$ experiments were conducted for the identified metabolites using an inclusion list with the accurate mass-to-charge ratio of the most abundant charge states. Here, the isolation window was set to $m/z = 2$, the resolution to 15,000 FWHM (at *m/z* 200), the scan range to *m/z* 200–1,500, and the NCE to 25%.

# 3 | RESULTS AND DISCUSSION

Besides protein-based inhibitors of MSTN and its signaling pathways, peptidic drug candidates have also been developed within the last years.[7–12] As none of these peptides have been tested in human subjects so far, there is currently no evidence for their efficacy as therapeutics for the treatment of muscular diseases and performance-enhancing agents in other species than mice. Nevertheless, the aim of this study was to develop a doping control detection assay for such MSTN inhibitory peptides from human serum. This is particularly important as their amino acid sequences are published and the peptides thus readily available through custom peptide synthesis. Moreover, it is a well-known fact that a lack of clinical tests/approval does not prevent athletes from misusing a promising drug candidate potentially offering performance-enhancing properties.

## 3.1 | Selection of target analytes

As mentioned above, two different natural regulators of MSTN were used as molecular basis for the development of peptidic MSTN inhibitors: the MSTN propeptide and FST. For the development and characterization of a doping control detection assay, the following drug candidates were selected as model compounds, and reference material was obtained by custom peptide synthesis.

19427611, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.3473 by Ohio State University Ohio Sta, Wiley Online Library on [04/04/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

### 3.1.1 | MSTN inhibitory peptides derived from the MSTN propeptide

Based on the research studies published by Takayama et al.[7–10] between 2015 and 2019, three different MSTN inhibitory peptides derived from the MSTN propeptide were selected as model compounds for this research project (Table 1). Peptide 7 comprises 23 amino acids (positions 21–43) of the murine MSTN propeptide and was found to represent the minimum sequence effectively inhibiting myostatin activity.[7] In Peptide 3d, the N-terminal tryptophan residue was replaced with a 2-naphthyloxyacetyl residue, and amino acid substitutions were introduced at positions 32 (Ala → Trp) and 38 (Leu → Ile) to increase the peptide's potency.[9] The resulting derivative Peptide 3d was found to have a considerably improved in vitro MSTN inhibitory activity, and significant increases in muscle mass by +10–35% and +10–19% were observed in wild-type and pathological mouse models (dosage: 2 × 30 nmol IM). To reduce antigenicity and production costs and to improve in vivo stability, the amino acid sequence of Peptide 3d was further modified by shortening, amino acid exchanges, and the introduction of the non-natural amino acids cyclohexylglycine (Chg) at two positions (Ile/Leu → Chg) and D-tryptophan at one position.[10] The resulting Peptide 8a/MIPE-1686 comprises 16 amino acids and was found to be twice as effective in in vitro MSTN inhibition as Peptide 3d. Moreover, in vivo studies could demonstrate that Peptide 8a/MIPE-1686 can increase not only muscle mass by 14 ± 6.6% but also muscle strength (approximately +30%) in a mouse model of muscular dystrophy (dosage: 2 × 30 nmol IM).

Even though the selected model peptides are derived from the murine MSTN propeptide, they were used for the development of a doping control assay applicable to human subjects. As the amino acid sequences of murine and human MSTN are 100% identical,[22] it can be assumed that the proposed MSTN inhibitory peptides are effective in both species. Moreover, Peptide 7 and the corresponding part of the human MSTN propeptide also share a high sequence homology of 91% (21/23 amino acids).[22] But as earlier studies have demonstrated a varying efficacy of other MSTN inhibitors between species,[5] further research is required to confirm the applicability of MSTN inhibitory peptides to humans.

Recently, Takayama et al.[23] presented MID-35, a MSTN inhibitory *retro-inverso* peptide comprising only D-amino acids as well as two amino acid exchanges at positions 5 and 8 (D-Ser → D-Arg; D-Gln → D-Arg), resulting in improved physicochemical properties and robustness against proteolytic degradation. In vitro, MID-35 showed a similar efficacy as peptide 8a/MIPE-1686, but its in vivo inhibitory potency was significantly improved due to the enhanced peptide stability. This peptide will be included in future studies.

### 3.1.2 | MSTN inhibitory peptides derived from FST

The glycoprotein FST is a natural inhibitor of MSTN and other TGF-β cytokines, and Saitoh et al.[11,12] used the amino acid sequence of its N-terminal domain (ND) responsible for MSTN binding to developed a series of MSTN inhibitory peptides. Within this research project, three different model peptides were used for method development and characterization: DF-3 comprises 14 amino acids of the FST N-terminal domain (ND) (positions 41–54), including three hydrophobic residues essential for MSTN binding, and represents the minimum core structure of human FST required for an effective in vitro and in vivo inhibition of the TGF-β cytokine.[11] In vivo, it was found to increase muscle mass by approximately 12% (dosage: 20 nmol IM). As the methionine residue at position 50 of FST is susceptible to oxidation, it was replaced with the non-natural amino acid 2-thienylalanine (Thi), resulting in model peptide DF-25.[12] In the third model peptide DF-100, the MSTN inhibitory activity could be further improved by replacing the hydrophobic residues valine ($Val_{41}$), tryptophan ($Trp_{49}$), $Thi_{50}$, and phenylalanine ($Phe_{52}$) with α-phenylglycine (Phg), Phe, homophenylalanine (Hph), and 4-fluorophenylalanine (Phe[4-F]). However, in vitro studies demonstrated that these FST-based peptides are significantly less effective in MSTN inhibition than the drug candidate derived from the MSTN propeptide, and further structural changes might be necessary.[12]

## 3.2 | Method development and optimization

### 3.2.1 | Sample preparation

During method development, different protocols for peptide extraction were tested, including protein precipitation alone as well as in combination with subsequent solid-phase extraction (SPE) (HR-XA [3 mL/60 mg, Macherey-Nagel, Düren, Germany], HR-XC [3 mL/60 mg, Macherey-Nagel, Düren, Germany], and HLB [OASIS, 3ccm/60 mg, Waters, Eschborn, Germany]). As protein precipitation without subsequent SPE was found to provide the highest recovery for all target peptides (data not shown), this approach was further optimized and finally characterized.

As shown in Table S1, most of the MSTN inhibitory peptides could only be detected when serum samples were acidified prior to protein precipitation. Both when a neutral pH of 6.5 (water) and basic pH of 9.0 (1% $NH_3$) were employed, Peptides 7, 3d, 8a, and DF-25 were not detectable, maybe due to the association with serum proteins. Regarding protein precipitation (Table S2), a volume of 400 µL (ratio between diluted sample and solvent of 1:2) was found to work best for Peptides 7, 3d, DF-3, and DF-25. For Peptide 8a and DF-100, volumes of 200 and 800 µL yielded slightly higher recoveries. Eventually, 400 µL were chosen as optimal volume to enable an efficient, simultaneous extraction of all target analytes.

### 3.2.2 | Mass spectrometric detection

To allow for an unambiguous mass spectrometric detection of the target peptides, two ion transitions were employed for each compound (Table 2). While the first precursor/product ion pair was evaluated for

19427611, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.3473 by Ohio State University Ohio Sta, Wiley Online Library on [04/04/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

analyte identification, the second one provides additional information for confirmation purposes. In Figures S1–S8, product ion mass spectra of the target analytes and both ISTDs are displayed, and the product ions selected for identification and confirmation purposes highlighted in blue and green. Exemplary extracted product ion chromatograms of a blank and a serum sample fortified with 20 ng/mL of the propeptide- and FST-based MSTN inhibitory peptides are shown in Figure 1. All analytes were unambiguously detected, and no interfering signals were observed in the blank specimen.

## 3.3 | Method characterization

To assess the applicability of the approach to sports drug testing, it was validated according to current WADA guidelines,[20] and results are summarized in Table 3. The method's specificity was investigated by analyzing a total of 10 different serum samples collected from healthy volunteers. As exemplarily shown in Figure 1, no interfering signals were present at the respective retention times (RT). For all target analytes, the approach was linear up to a concentration of 100 ng/mL as reflected by correlation coefficients ($r^2$) higher than 0.97. Thereby, the linearity determined for the peptides derived from the MSTN propeptide was found to be better than that for the FST-based analytes ($r^2 \geq 0.994$ vs. $r^2 \geq 0.974$). For the estimation of the detection limits, a detection rate of at least 95% was applied to six serum samples fortified with the four lowest analyte concentrations detectable during linearity measurements (2, 5, 10, and 20 ng/mL), and the estimated LODs for the six MSTN inhibitory peptides ranged from 3 to 9 ng/mL. Considering the therapeutic doses used in animal studies (e.g., Peptide 8a/MIPE-1686: 2 × 30 nmol (=67.6 µg) IM/20 g mouse → ~3.4 mg/kg[10]), the achieved detection limits should theoretically be adequate for human administration scenarios. But since pharmacokinetic data for MSTN inhibitory peptides are currently not available and the past has additionally shown that the effectiveness—and thus dosage—of MSTN inhibitors can strongly vary between species,[5] authentic human and/or animal post-administration samples are required to finally assess method sensitivity. The identification capability was successfully demonstrated by analyzing 10 different serum specimens fortified with 20 ng/mL of the target peptides. Here, a high inter-individual variability was observed (data not shown), which can probably be attributed to differences in peptide stability/metabolism (see also below). Two concentration levels (20 and 40 ng/mL) were employed to determine the method's recovery, which was found to vary from 20% to 48% for most of the analytes. However, Peptide 8a was characterized by a significantly lower recovery of only 8%. Possible explanations for this phenomenon could be metabolic degradation and losses during sample preparation, for example, due to the association with serum proteins or nonspecific binding to the sample tubes. In a study published by Takayama et al.,[24] the enzymatic stability of different MSTN inhibitory peptides derived from the MSTN propeptide was systematically investigated. Peptide 8a/MIPE-1686 was found to be very robust towards degradation when incubated at 37°C with three different proteases, but nonspecific binding to experimental instruments could be observed, resulting in losses of up to 15%. Consequently, the most likely explanation for the poor recovery is a combination of (non-)specific binding to serum proteins and sample tubes. Sample carryover was below 5%, and the experiment investigating potential matrix effects indicated analyte loss in

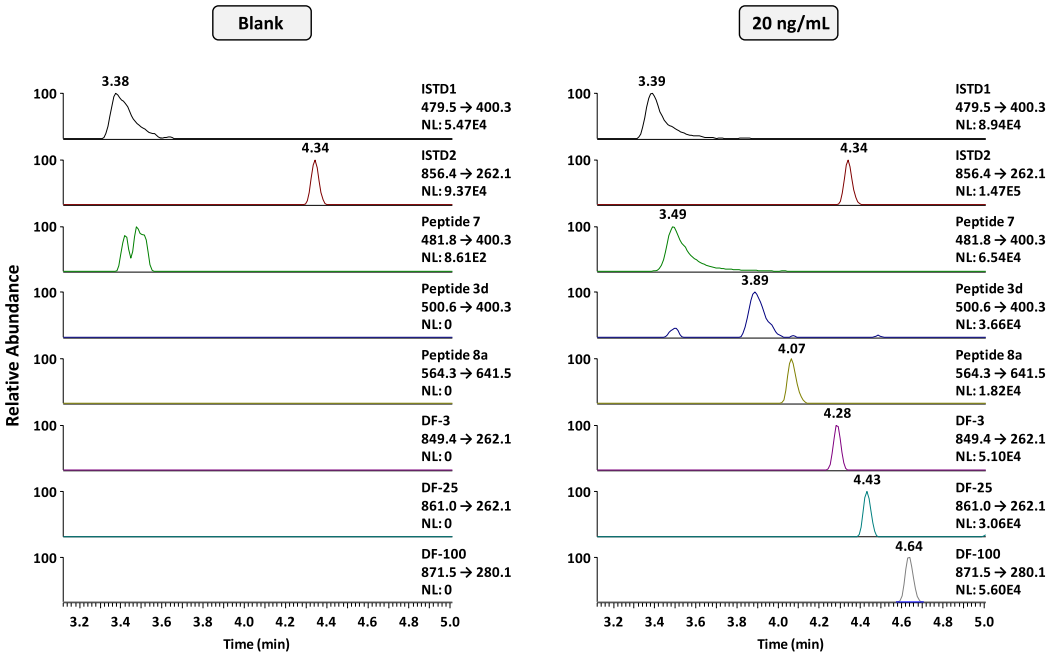


**FIGURE 1** Exemplary extracted product ion chromatograms of a blank and a serum sample fortified with 20 ng/mL of the different MSTN inhibitory peptides. Samples were measured on an Orbitrap Exploris™ 480 mass spectrometer coupled to a Vanquish™ UHPLC. [Colour figure can be viewed at wileyonlinelibrary.com]

19427611, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.3473 by Ohio State University Ohio Sta, Wiley Online Library on [04/04/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**TABLE 3** Results of method validation.

| Validation parameter | *n* | Concentration(s): | Target analyte | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | Peptide 7 | Peptide 3d | Peptide 8a | DF-3 | DF-25 | DF-100 |
| Specificity | 10 | - | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Robustness (Plasma) | 10 | 20 ng/mL | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Linearity | 11 | 1–100 ng/mL | $R^2 = 0.998$ | $R^2 = 0.995$ | $R^2 = 0.994$ | $R^2 = 0.980$ | $R^2 = 0.974$ | $R^2 = 0.980$ |
| LOD | 6 | 2–20 ng/mL | 3 ng/mL | 4 ng/mL | 9 ng/mL | 4 ng/mL | 9 ng/mL | 7 ng/mL |
| Recovery | 6 | 20 ng/mL | 30.5% | 38.4% | 7.7% | 33.9% | 35.7% | 27.4% |
| | 6 | 40 ng/mL | 31.4% | 36.2% | 8.4% | 20.7% | 22.5% | 48.2% |
| Identification capability | 10 | 20 ng/mL | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Carryover | - | 80 ng/mL | <5% | <5% | <5% | - | - | - |
| Matrix effects | - | 20 ng/mL | 290% | 303% | 133% | 172% | 241% | 322% |
| Sample stability at RT | - | 20 ng/mL | <1 h | ≥24 h | ≥14 d | <1 h | <1 h | ≥8 h |
| Sample stability at 4°C | - | 20 ng/mL | ≥4 h | ≥14 d | ≥14 d | <2 h | <2 h | ≥7 d |
| Extract stability at 10°C | 10 | 20 ng/mL | ≥140 h | ≥140 h | ≥140 h | ≥140 h | ≥140 h | ≥140 h |

water—probably due to an adsorption to the surface of the LC vials—as the detected signals were significantly higher when serum extracts were used. To determine the robustness of the approach, plasma instead of serum was used as biological matrix and an analyte concentration of 20 ng/mL was successfully detected in 10 different samples.

To maintain sample integrity and ensure maximum levels of quality for sample analysis, very strict rules apply to the collection, transportation, and storage of doping control blood samples[25]: They have to be transported in refrigerated condition in a certain period of time, and the temperature in the transportation device is continuously monitored by a data logger until sample reception. Within this study, the stability of the different MSTN inhibitory peptides was therefore assessed both at RT and 4°C. For these experiments, a concentration of 20 ng/mL was chosen as all MSTN inhibitory peptides are administered by intramuscular injection, which probably results in comparably low blood levels of these drugs. At RT, Peptide 7, DF-3, and DF-25 were found to be rapidly degraded within 1 h. By contrast, DF-100 and Peptide 3d could be detected for 8 and 24 h. Peptide 8a was clearly detectable for the entire 2 weeks of incubation, indicating a high resistance towards enzymatic degradation. When serum samples were stored refrigerated, peptides DF-3 and DF-25 were also degraded within a very short time. Peptide 7 was found to be detectable for 4 h and the remaining target compounds DF-100, Peptide 3d, and Peptide 8a for 1 and 2 weeks. The extended detection windows for some of the analytes can probably be attributed to the lower activity of proteolytic enzymes at 4°C. Overall, these results are in accordance with the literature,[24] where the resistance of propeptide-derived MSTN inhibitors towards exopeptidases and endopeptidases was studied, and Peptide 8a/MIPE-1686 was found to be the most stable one. Regarding sports drug testing, the low stability of peptides DF-3, DF-25, and Peptide 7 both at RT and 4°C could be a practical issue complicating their detection in doping control specimens due to degradation during sample transportation. However, authentic post-administration samples are required to finally assess whether drug doses, serum levels, analyte stability, and method sensitivity are compatible. Moreover, additional experiments investigating analyte stability in (dried) whole blood and/or plasma should be conducted, as there can be significant differences between biological matrices (see also Section 3.4).

Additionally, analyte stability in serum sample extracts (analyte concentration: 20 ng/mL) kept for 140 h at an autosampler temperature of 10°C was studied, and all target peptides were found to be detectable for the entire period of time.

## 3.4 | Peptide metabolism

During method development and optimization, both for the FST-derived MSTN inhibitory peptide DF-3 ($m/z = 849.42$) and the corresponding ISTD2 ($m/z = 856.43$), oxidation of the sulfur-containing amino acid methionine was observed. As shown in Figure 2, the presence of methionine sulfoxide resulted in mass-to-charge ratios of 857.42 and 864.43, and a RT shift of approximately −0.3 min. The corresponding product ion mass spectra are displayed in Figure S9. As the high susceptibility of methionine towards oxidation is a well-known phenomenon that can have significant impact on the effectiveness of protein pharmaceuticals,[26] the methionine residue present in DF-3 was replaced with 2-thienylalanine (Thi) in DF-25 and homophenylalanine (Hph) in DF-100, which were found to have a comparable or even better in vitro inhibitory activity.[12] In sports drug testing assays, both unmodified and oxidized DF-3 should be implemented, as the latter appeared to be a highly abundant derivative.

In general, blood represents one of the main sites for the metabolism/degradation of therapeutic proteins and peptides.[27] In order to identify potential metabolites and degradation products, the different MSTN inhibitory peptides were separately incubated for 2 h in serum (concentration: 250 ng/mL) and subsequently subjected to sample

19427611, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.3473 by Ohio State University Ohio Sta, Wiley Online Library on [04/04/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

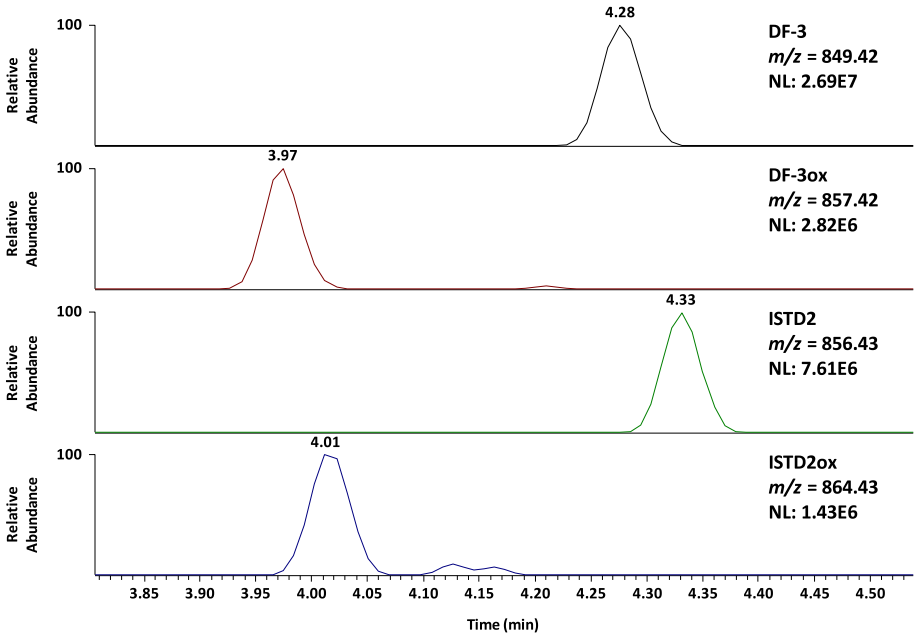


**FIGURE 2** Extracted ion chromatograms of peptide DF-3 and ISTD2 as well as their oxidized derivatives. The corresponding serum sample was fortified with 100 ng/mL of the target analytes and 25 ng/mL of the ISTDs and analyzed on an Orbitrap Exploris™ 480 mass spectrometer coupled to a Vanquish™ UHPLC. [Colour figure can be viewed at wileyonlinelibrary.com]

**TABLE 4** Summary of the metabolites identified for the different MSTN inhibitory peptides.

| Name | Amino acid sequence | Monoisotopic mass (Da) | *m/z* | *z* | RT |
|---|---|---|---|---|---|
| **Peptide 7** | WRQNTRYSRIEAIKIQILSKLRL | 2,883.70 | 481.62 | 6 | 3.51 |
| M1 | YSRIEAIKIQILSKLRL | 2,042.27 | 511.58 | 4 | 3.64 |
| M2 | SRIEAIKIQILSKLRL | 1,879.21 | 470.81 | 4 | 3.54 |
| M3 | IEAIKIQILSKLRL | 1,636.08 | 546.37 | 3 | 3.57 |
| M4 | EAIKIQILSKLRL | 1,522.99 | 508.67 | 3 | 3.46 |
| **Peptide 3d** | [X]RQNTRYSRIEWIKIQIISKLRL | 2,996.72 | 500.46 | 6 | 3.89 |
| M1 | YSRIEWIKIQIISKLRL | 2,157.32 | 540.34 | 4 | 3.76 |
| M2 | IEWIKIQIISKLRL | 1,751.12 | 584.71 | 3 | 3.77 |
| M3 | IEWIKIQIISKL | 1,482.92 | 742.47 | 2 | 3.95 |
| **Peptide 8a** | WYIR[D-TRP]IK[Chg]QIWSK[Chg]RL | 2,252.35 | 564.09 | 4 | 4.08 |
| **DF-3** | VNDNTLFKWMIFNG | 1,696.84 | 849.42 | 2 | 4.27 |
| M1 | NDNTLFKWMIFNG | 1,597.77 | 799.89 | 2 | 4.24 |
| **DF-25** | VNDNTLFKW[Thi]IFNG | 1,718.82 | 860.42 | 2 | 4.41 |
| M1 | NDNTLFKW[Thi]IFNG | 1,619.76 | 810.89 | 2 | 4.39 |
| **DF-100** | [Phg]NDNTLFKF[Hph]I[Phe(4-F)]NG | 1,739.87 | 870.93 | 2 | 4.62 |

extraction and untargeted LC-MS/MS analysis. Identified metabolites are summarized in Table 4 and extracted ion chromatograms shown in Figure 3. As to be expected, the number of detected derivatives decreased with the degree of peptide modification as the inclusion of non-natural amino acids can improve peptide stability and proteolytic resistance.[27] For Peptides 7 and 3d, several shorter metabolites lacking 6–10 amino acids at the N-terminus were identified. The MS/MS spectra of these derivatives are depicted in Figures S10–S16. Compared to the intact target peptides, the relative abundances of these metabolites were lower except for M3 of Peptide 7, which could therefore also be a valuable target for sports drug testing. By contrast, no metabolites could be detected for Peptide 8a/MIPE-1686. As already mentioned above, a study conducted by Takayama et al.[24] has demonstrated that Peptide 8a/MIPE-1686 has a high resistance towards exo- and endopeptidases, which could also be confirmed within this research project. For DF-3 and DF-25, highly abundant derivatives lacking the N-terminal valine residue were identified, and the corresponding product ion mass spectra of DF-3 M1 and DF-25 M1 are shown in Figures 4 and 5. As their relative abundances were found to be higher than those of the intact peptides (Figure 3), they should also be implemented into the doping control detection method. For DF-100, no derivatives were detected, which can probably be attributed to the presence of several non-natural amino acids. To confirm the in vivo formation of these metabolites, authentic

19427611, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.3473 by Ohio State University Ohio Sta, Wiley Online Library on [04/04/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

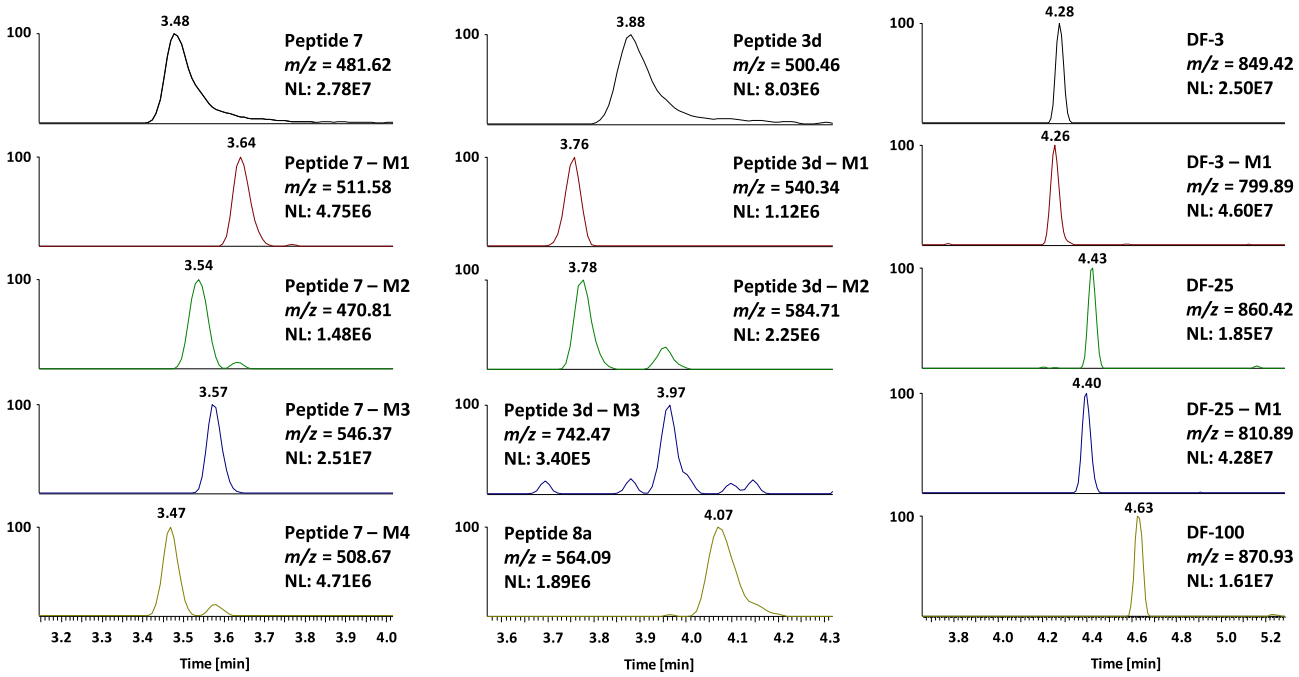


**FIGURE 3** Extracted ion chromatograms of the target analytes and their metabolites. The depicted serum sample was fortified with 100 ng/mL of the different MSTN inhibitory peptides and 25 ng/mL of the ISTDs and analyzed on an Orbitrap Exploris™ 480 mass spectrometer coupled to a Vanquish™ UHPLC. [Colour figure can be viewed at wileyonlinelibrary.com]

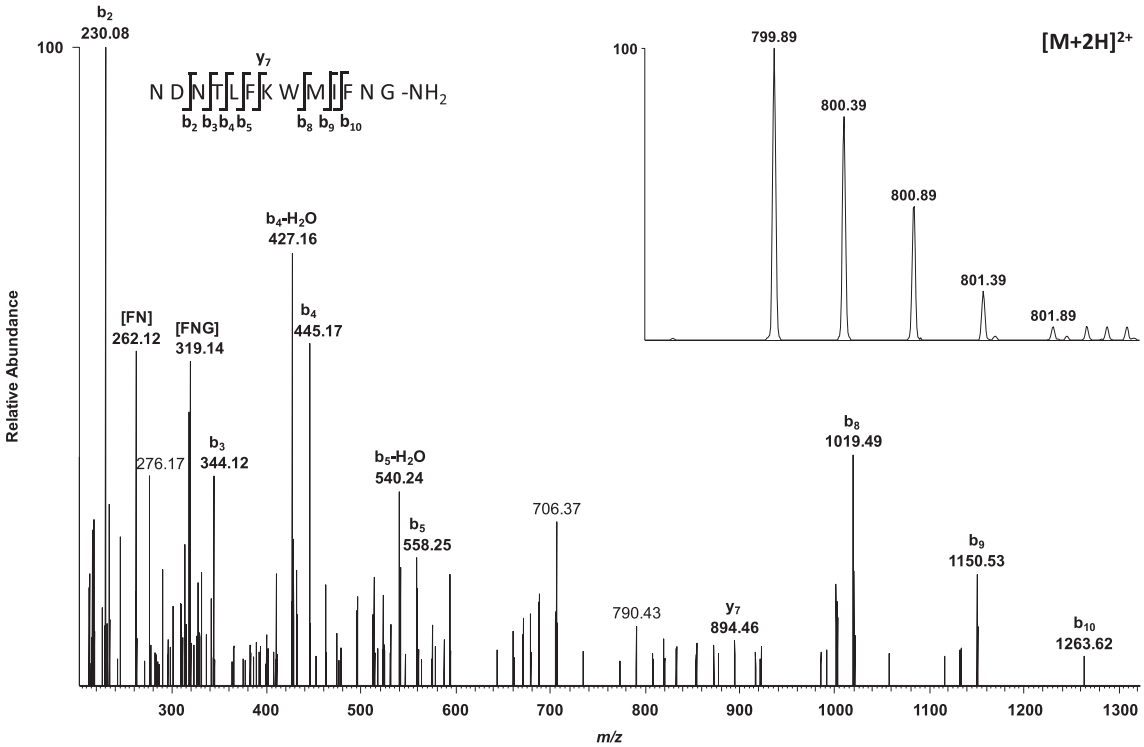


**FIGURE 4** Product ion spectrum of DF-3—M1 ($m/z = 849.42$) recorded on an Orbitrap Exploris™ 480 mass spectrometer interfaced with a Vanquish™ UHPLC.

19427611, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.3473 by Ohio State University Ohio Sta, Wiley Online Library on [04/04/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

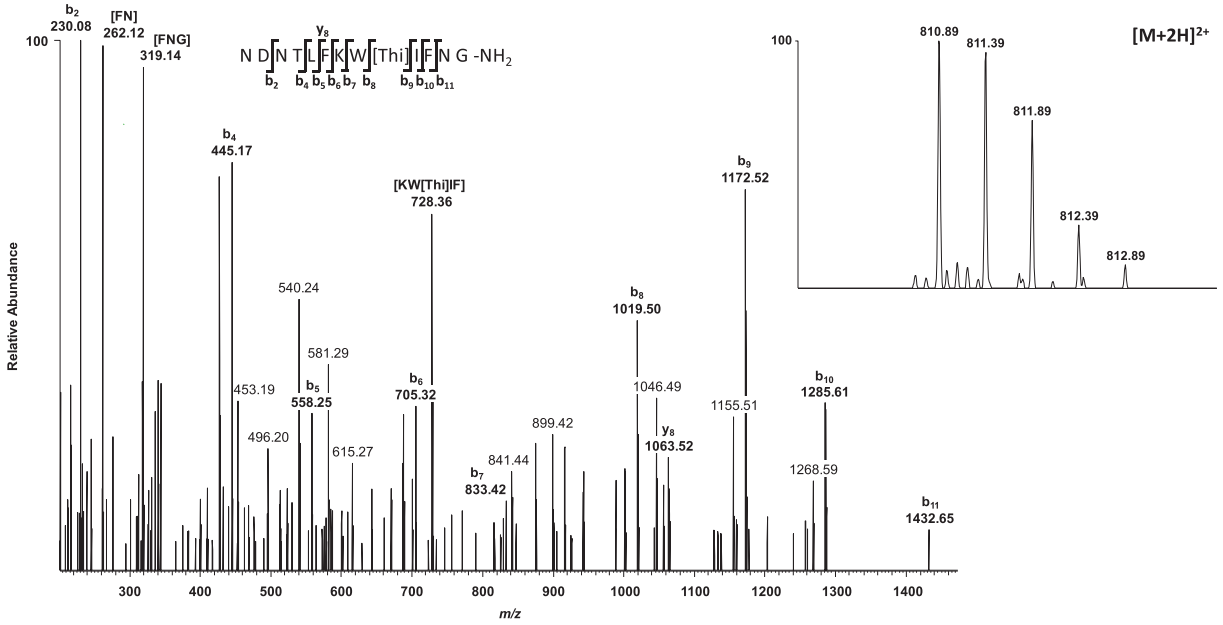


**FIGURE 5** Product ion spectrum of DF-25—M1 ($m/z = 810.89$) recorded on an Orbitrap Exploris™ 480 mass spectrometer interfaced with a vanquish™ UHPLC.

human post-administration samples are required. But as all MSTN inhibitory peptides are still in pre-clinical development, the availability of such specimens is scarce.

Finally, the Full MS data acquired during the stability experiments were retrospectively evaluated to determine the stability of the metabolites M3 of Peptide 7 and M1 of DF-3 and DF-25. Unfortunately, these derivatives were also found to be characterized by a poor stability and could not improve the detection windows of these MSTN inhibitory peptides. But as these derivatives could already be detected in the serum samples extracted without any incubation ($T_0$), the degradation of these peptides appears to take place rather quickly.

## 4 | CONCLUSIONS

Inhibitors of the TGF-β cytokine MSTN and its signaling pathways can significantly increase skeletal muscle mass and are therefore both promising drug candidates for the therapy of muscle wasting disorders and doping agents in sports. While several doping control detection methods for protein-based MSTN inhibitors have been proactively developed, there is currently no such approach for MSTN inhibitory peptides. Within this study, protein precipitation and LC-HRMS/MS were employed to develop a detection method for six peptidic model compounds derived from the MSTN propeptide and FST, which are important natural regulators of MSTN. While the presented approach was found to be specific, sensitive, and robust, the stability of some analytes (Peptide 7, DF-3, and DF-25) in serum was found to be a critical factor, which can potentially interfere with sports drug testing routine procedures. For all three peptides, abundant metabolites were identified, which were characterized by an equally limited stability in serum. As mentioned above, additional experiments on the stability in whole blood and/or plasma should be conducted in the near future. For example, the use of dried blood spots (DBS) could represent a promising alternative, as peptidic doping agents such as Synacthen were found to be significantly more stable in this complementary biological matrix than in conventional serum samples.[28] Moreover, authentic post-administration samples and custom-synthesized reference material are required to confirm the in vivo formation of the proposed metabolites. As the analysis of such specimens would also be important as proof-of-concept, an animal administration study could represent a valuable alternative.

Even though future studies still need to confirm the general effectiveness of such peptides in humans and there are a few others questions left, the data acquired within this research project show that at least the probably most relevant MSTN inhibitory peptides Peptide 8a/MIPE-1686 and DF-100 can be readily implemented into doping control routine analysis, as they were found to be stable in serum for 2 and 1 weeks, which should be compatible with routine processes regarding sample transportation and storage.

The number of potential therapeutics aiming at the blockade of MSTN and/or the activin receptor signaling pathways is continuously increasing and includes not only different protein-/peptide-derived inhibitors[4] but also nucleic acid-based agents such as antisense oligonucleotides.[29] Even though probably only a few drug candidates will eventually gain clinical approval, it is of utmost importance for preventive doping research to continuously expand the range of available tests to deter athletes from misusing such substances.

19427611, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.3473 by Ohio State University Ohio Sta, Wiley Online Library on [04/04/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

## ACKNOWLEDGEMENTS

The presented work was conducted with support of the National Anti-Doping Agency (NADA) Germany; the Federal Ministry of the Interior, Building, and Community of the Federal Republic of Germany; the ERASMUS+ Programme of the European Commission; and the Manfred-Donike Institute for Doping Analysis (Cologne, Germany). Open Access funding enabled and organized by Projekt DEAL.

## ORCID

*Katja Walpurgis* https://orcid.org/0000-0003-1194-8158
*Andreas Thomas* https://orcid.org/0000-0003-1199-0743
*Mario Thevis* https://orcid.org/0000-0001-5972-8854

## REFERENCES

1. McPherron AC, Lawler AM, Lee SJ. Regulation of skeletal muscle mass in mice by a new TGF-beta superfamily member. *Nature*. 1997; 387(6628):83-90. doi:10.1038/387083a0
2. Lee SJ, McPherron AC. Regulation of myostatin activity and muscle growth. *Proc Natl Acad Sci U S A*. 2001;98(16):9306-9311. doi:10.1073/pnas.151270098
3. Lee SJ. Regulation of muscle mass by myostatin. *Annu Rev Cell Dev Biol*. 2004;20(1):61-86. doi:10.1146/annurev.cellbio.20.012103.135836
4. Suh J, Lee YS. Myostatin inhibitors: panacea or predicament for musculoskeletal disorders? *J Bone Metab*. 2020;27(3):151-165. doi:10.11005/jbm.2020.27.3.151
5. Latres E, Mastaitis J, Fury W, et al. Activin A more prominently regulates muscle mass in primates than does GDF8. *Nat Commun*. 2017;8(1):15153. doi:10.1038/ncomms15153
6. Campbell C, McMillan HJ, Mah JK, et al. Myostatin inhibitor ACE-031 treatment of ambulatory boys with Duchenne muscular dystrophy: results of a randomized, placebo-controlled clinical trial. *Muscle Nerve*. 2017;55(4):458-464. doi:10.1002/mus.25268
7. Takayama K, Noguchi Y, Aoki S, et al. Identification of the minimum peptide from mouse myostatin prodomain for human myostatin inhibition. *J Med Chem*. 2015;58(3):1544-1549. doi:10.1021/jm501170d
8. Takayama K, Nakamura A, Rentier C, et al. Effect of N-terminal acylation on the activity of myostatin inhibitory peptides. *ChemMedChem*. 2016;11(8):845-849. doi:10.1002/cmdc.201500533
9. Takayama K, Rentier C, Asari T, et al. Development of potent myostatin inhibitory peptides through hydrophobic residue-directed structural modification. *ACS Med Chem Lett*. 2017;8(7):751-756. doi:10.1021/acsmedchemlett.7b00168
10. Takayama K, Asari T, Saitoh M, et al. Chain-shortened myostatin inhibitory peptides improve grip strength in mice. *ACS Med Chem Lett*. 2019;10(6):985-990. doi:10.1021/acsmedchemlett.9b00174
11. Saitoh M, Takayama K, Hitachi K, et al. Discovery of a follistatin-derived myostatin inhibitory peptide. *Bioorg Med Chem Lett*. 2020; 30(3):126892. doi:10.1016/j.bmcl.2019.126892
12. Saitoh M, Takayama K, Roppongi Y, et al. Strategic structure-activity relationship study on a follistatin-derived myostatin inhibitory peptide. *Bioorg Med Chem Lett*. 2021;46:128163. doi:10.1016/j.bmcl.2021.128163
13. World Anti-Doping Agency (WADA). World Anti-Doping Code—international standard: Prohibited List 2023. Available at: www.wada-ama.org. 2022.
14. Walpurgis K, Thomas A, Schänzer W, Thevis M. Myostatin inhibitors in sports drug testing: detection of myostatin-neutralizing antibodies in plasma/serum by affinity purification and Western blotting. *Proteomics Clin Appl*. 2016;10(2):195-205. doi:10.1002/prca.201500043
15. Walpurgis K, Thomas A, Thevis M. Detection of the myostatin-neutralizing antibody Domagrozumab in serum by means of Western blotting and LC-HRMS. *Drug Test Anal*. 2019;11(11-12):1714-1723. doi:10.1002/dta.2729
16. Walpurgis K, Thomas A, Dellanna F, Schänzer W, Thevis M. Detection of the human anti-ActRII antibody bimagrumab in serum by means of affinity purification, tryptic digestion, and LC-HRMS. *Proteomics Clin Appl*. 2018;12(3):e1700120. doi:10.1002/prca.201700120
17. Reichel C, Gmeiner G, Thevis M. Detection of black market follistatin 344. *Drug Test Anal*. 2019;11(11-12):1675-1697. doi:10.1002/dta.2741
18. Walpurgis K, Weigand T, Knoop A, et al. Detection of follistatin-based inhibitors of the TGF-beta signaling pathways in serum/plasma by means of LC-HRMS/MS and Western blotting. *Drug Test Anal*. 2020; 12(11-12):1636-1648. doi:10.1002/dta.2925
19. Thomas A, Schanzer W, Thevis M. Determination of human insulin and its analogues in human blood using liquid chromatography coupled to ion mobility mass spectrometry (LC-IM-MS). *Drug Test Anal*. 2014;6(11-12):1125-1132. doi:10.1002/dta.1710
20. World Anti-Doping Agency (WADA). World Anti-Doping Code—international standard: Laboratories 2021. Available at: www.wada-ama.org. 2020.
21. Thomas A, Knoop A, Schanzer W, Thevis M. Characterization of in vitro generated metabolites of selected peptides <2 kDa prohibited in sports. *Drug Test Anal*. 2017;9(11-12):1799-1803. doi:10.1002/dta.2306
22. National Library of Medicine (NLM). BLAST blastp suite. Available at: https://blast.ncbi.nlm.nih.gov/Blastcgi
23. Takayama K, Hitachi K, Okamoto H, et al. Development of myostatin inhibitory d-peptides to enhance the potency, increasing skeletal muscle mass in mice. *ACS Med Chem Lett*. 2022;13(3):492-498. doi:10.1021/acsmedchemlett.1c00705
24. Takayama K, Odagiri M, Taguchi A, Taniguchi A, Hayashi Y. Enzymatic stability of myostatin inhibitory 16-mer peptides. *Chem Pharm Bull(Tokyo)*. 2020;68(6):512-515. doi:10.1248/cpb.c20-00158
25. World Anti-Doping Agency (WADA). World Anti-Doping Code—international standard: Testing and Investigations 2021 Available at: www.wada-ama.org. 2019.
26. Li S, Schoneich C, Borchardt RT. Chemical instability of protein pharmaceuticals: mechanisms of oxidation and strategies for stabilization. *Biotechnol Bioeng*. 1995;48(5):490-500. doi:10.1002/bit.260480511
27. Böttger R, Hoffmann R, Knappe D. Differential stability of therapeutic peptides with different proteolytic cleavage sites in blood, plasma and serum. *PLoS ONE*. 2017;12(6):e0178943. doi:10.1371/journal.pone.0178943
28. Tretzel L, Thomas A, Geyer H, Delahaut P, Schänzer W, Thevis M. Determination of Synacthen((R)) in dried blood spots for doping control analysis using liquid chromatography tandem mass spectrometry. *Anal Bioanal Chem*. 2015;407(16):4709-4720. doi:10.1007/s00216-015-8674-6
29. Maeta K, Farea M, Nishio H, Matsuo M. An antisense oligonucleotide against a splicing enhancer sequence within exon 1 of the MSTN gene inhibits pre-mRNA maturation to act as a novel myostatin inhibitor. *Int J Mol Sci*. 2022;23(9):5016. doi:10.3390/ijms23095016

## SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

**How to cite this article:** Walpurgis K, Agricola J, Thomas A, Thevis M. Myostatin inhibitory peptides in sports drug testing. *Drug Test Anal*. 2023;1-11. doi:10.1002/dta.3473

19427611, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.3473 by Ohio State University Ohio Sta, Wiley Online Library on [04/04/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License